

P01 Point-Cloud Stockpile Height and Volume Analysis

2024 USGS LiDAR top surface compared with an interpolated local base surface

Height above base was calculated by subtracting a sloping TIN base surface from a 1 ft point-cloud-derived top surface. The base was interpolated from eight surrounding floor-control samples.

Preferred volume estimate: 4,237 yd³

Constant-floor comparison: 4,361 yd³

Sensitivity difference: +2.93%

Maximum height above base: 30.85 ft

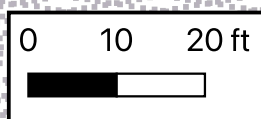
Mean point density: 0.88 points/ft²
(9.5 points/m²)

P01 classification review:
70.41% Ground
29.59% Unclassified

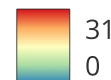
The supplied ground class included much of the stockpile surface, so Class 2 points were reviewed but not used directly as the base.

A-A: longitudinal profile
B-B: transverse profile

Volume is an analytical estimate dependent on boundary interpretation, rasterization, and base-surface assumptions.



Height above base (ft)



Horizontal CRS: NAD83(2011) / Florida East (ftUS), EPSG:6438
Vertical datum: NAVD88 height (ftUS), EPSG:6360
Source: USGS 3DEP LiDAR, Miami-Dade County, 2024